

Domain and Range

Problem-1: Find the domain and range for the function $f(x) = x^2 - x + 1$, where $x = \{0, 1, 2, 3\}$.

Solution:

When $x = 0$, $f(0) = 1$,

$x = 1$, $f(1) = 1$,

$x = 2$, $f(2) = 3$,

$x = 3$, $f(3) = 7$,

$\therefore \text{dom } f = \{0, 1, 2, 3\}$ and

$\text{range } f = \{1, 3, 7\}$.

Problem-2: Find the domain and range for the function, $f(x) = \frac{1}{x-3}$.

Solution:

Clearly, $f(x) = \frac{1}{x-3}$ is not defined when $x = 3$.

$\therefore \text{dom } f(x) = R - \{3\} = \text{set of all real numbers except } 3$.

Let $y = f(x) = \frac{1}{x-3}$

or, $x - 3 = \frac{1}{y}$

or, $x = \frac{1}{y} + 3 = \frac{1+3y}{y}$

So, x is not defined for $y = 0$.

$\therefore \text{range } f(x) = R - \{0\} = \text{set of all real numbers except } 0$.

Problem-3: Find the domain and range for the function, $f(x) = \frac{1+x}{5-x}$.

Solution:

Clearly, $f(x) = \frac{1+x}{5-x}$ is not defined when $x = 5$.

$\therefore \text{dom } f(x) = R - \{5\} = \text{set of all real numbers except } 5$.

Let $y = f(x) = \frac{1+x}{5-x}$ ○

or, $1+x = 5y - xy$

or, $x + xy = 5y - 1$

or, $x(1+y) = 5y - 1$

$\therefore x = \frac{5y-1}{1+y}$

So, x is not defined for $y = -1$.

$\therefore \text{range } f(x) = R - \{-1\} = \text{set of all real numbers except } -1$.

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Problem-4: Find the domain and range for the function, $f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$.

Solution:

Now, $f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$.

or, $f(x) = \frac{(x-2)(x-1)}{(x+3)(x-2)}$

So, $f(x)$ is not defined for $x = 2$ or $x = -3$.

$\therefore \text{dom } f(x) = R - \{2, -3\}$ = set of all real numbers except 2 and -3.

Let $y = f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$

or, $x^2y + xy - 6y = x^2 - 3x + 2$

or, $x^2(1-y) - x(3+y) + 6y + 2 = 0$

or, $x = \frac{(y+3) \pm \sqrt{(y+3)^2 - 4(1-y)(6y+2)}}{2(1-y)}$

$$[\because \text{If } ax^2 + bx + c = 0, a \neq 0 \text{ then } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}]$$

So, x is not defined for $y = 1$.

$\therefore \text{range } f(x) = R - \{1\}$ = set of all real numbers except 1.

H.W: Problem-5: Determine whether y is a function of x for $x^2 + y^2 = 4$.

Domain and ranges of some functions:

Function	Domain	Range
x	$(-\infty, \infty)$ or R	$(-\infty, \infty)$ or R
x^2	$(-\infty, \infty)$ or R	$[0, \infty)$
x^3	$(-\infty, \infty)$ or R	$(-\infty, \infty)$ or R
$\frac{1}{x}$	$R - \{0\}$	$R - \{0\}$
$\sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$\sqrt{x+2}$	$[-2, \infty)$	$[0, \infty)$
$\frac{1}{\sqrt{x+2}}$	$(-2, \infty)$	$(0, \infty)$
$\sin(x)$	$(-\infty, \infty)$ or R	$[-1, 1]$
$\cos(x)$	$(-\infty, \infty)$ or R	$[-1, 1]$
$\sin^2(x)$	$(-\infty, \infty)$ or R	$[0, 1]$
$\sin(x^2)$	$(-\infty, \infty)$ or R	$[-1, 1]$
$\ln(x)$	$(0, \infty)$	$(-\infty, \infty)$ or R
e^x	$(-\infty, \infty)$ or R	$(0, \infty)$